1 Capture, on the device

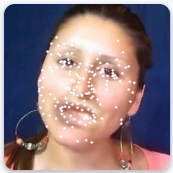
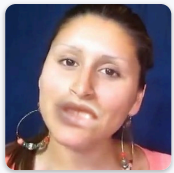


photo with its 478 landmarks

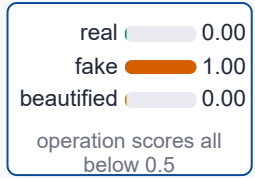


aligned crop, as in training

2 Detect



backbone

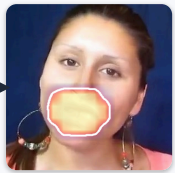


heads: class scores, operation scores



evidence map 16 × 16

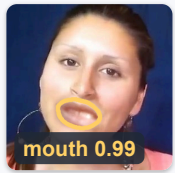
3 Locate, by code



map on the face, contour at 0.5



five part masks from landmarks



named part: most evidence

4 Explain and check, by code

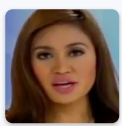
The model's decision is **fake** (score **1.00**). The evidence is concentrated on the **mouth** (mean evidence **0.99**). The score is the model's raw output, not a calibrated probability.

✓ H1–H6 match the record

✓ G7–G9 no score misuse

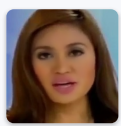
Training, once, offline

genuine



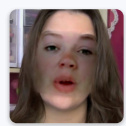
as recorded

+



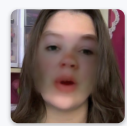
smoothed view

forged



as recorded

+



smoothed view

Skin smoothing is shown to both classes, so it cannot become the forgery cue. Genuine: FF++ frames and FFHQ originals. Forged: FF++ forgeries and 61,510 part-level edits with exact footprints (the evidence head's targets). Beautified: renders of two retouching services.

neural network · code with fixed rules · checks · steps 1–4 run in the browser